

Lecture 21: A note on mixed derivatives for unconstrained opt.
21/4/23

Quick revision of method of Lagrange multipliers
Points of max/min on a closed bounded domain

Bonus: Proof for Lagrange multipliers for fns of
2 variables.

If a function is of class C^2 , all its partial derivatives up to order 2. (i.e. $f_{xx}, f_{xy}, f_{yx}, f_{yy}$) are all continuous.

Clairaut-Schwartz tells us if f_{xy} and f_{yx} exist around (a,b) and are continuous at (a,b) , then $f_{xy}(a,b) = f_{yx}(a,b)$.
Thus, if a function is of class C^2 , $f_{xy}(x,y) = f_{yx}(x,y)$
 $\forall (x,y) \in \text{Domain}$.

If you look back at theorem 0.5 in max-min-I, you'll see that we only define $D(a,b)$ and the subsequent test for the nature of a critical point, if f is of class C^2 .
Thus, we'll always give you a function for unconstrained optimization that's of class C^2 , so you may assume $f_{xy} = f_{yx}$ ✓

By the way, you can use this fact to make your life slightly easier, in some case. Consider f_x and f_y , and decide if it's easier to find f_{yx} or f_{xy} .

E.g. from tut 8: $f(x,y) = x^2 - \frac{4xy}{y^2+1}$

$$f_x = 2x - \frac{4y}{y^2+1}$$

$$f_y = -\frac{4x(1-y^2)}{(y^2+1)^2}$$

notice $\frac{\partial}{\partial y}(f_x)$ requires quotient rule

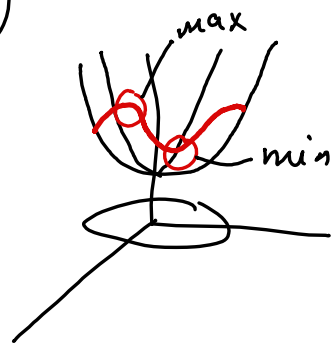
$\frac{\partial}{\partial x}(f_y)$ is easy!

$$f_{xy} = -\frac{4(1-y^2)}{(y^2+1)^2}$$

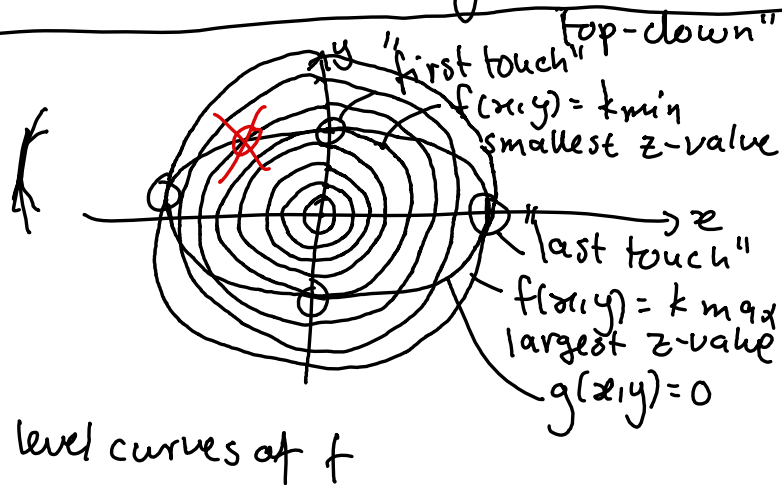
method of lagrange multipliers:

"find the points of max/min of $f(x,y)$ subject to $g(x,y)=0$ "

Solve for $(x,y), \lambda$ s.t.
$$\begin{cases} f_x(x,y) = \lambda g_x(x,y) \\ f_y(x,y) = \lambda g_y(x,y) \\ g(x,y) = 0 \end{cases}$$



evaluate $f(x,y)$ for all the (x,y) you found, to determine which give max or min.



notice, at point of min/max, tangent of $g(x,y)=0$ and tangent of level curve of $f(x,y)$ are parallel. If tangents are parallel, $\nabla g(a,b) \parallel \nabla f(a,b)$ since $\nabla g \perp \text{tangent}$

Find the points of min/max of $f(x,y,z)$ subject to $h(x,y,z)=0$

Solve for $(x,y,z), \lambda$ s.t.

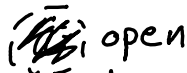
$$\begin{cases} f_x(x,y,z) = \lambda h_x(x,y,z) \\ f_y(x,y,z) = \lambda h_y(x,y,z) \\ f_z(x,y,z) = \lambda h_z(x,y,z) \\ h(x,y,z) = 0 \end{cases}$$

Evaluate $f(x,y,z)$ for all the (x,y,z) you found to determine max/min.

We seek the points of max/min for a continuously differentiable $f: \mathbb{R}^n \rightarrow \mathbb{R}$ on a closed bounded domain D in $\mathbb{R}^n$.
(we'll use $n=2$ or $n=3$)



closed



open

includes its own boundary

"finite"
"can draw all of it"

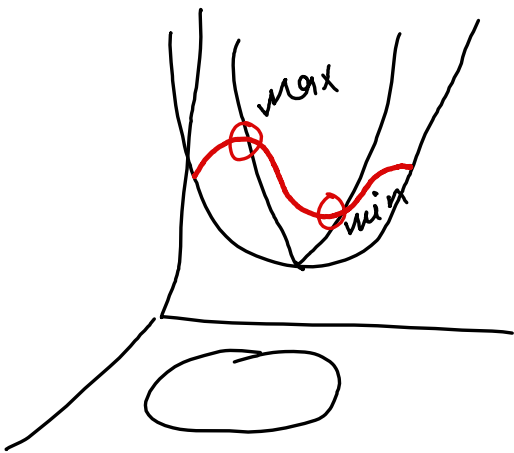
Note that the existence of such points follows from a higher-dimensional version of the extreme value theorem: we will find pt of max and pt of min.

In contrast, when you have an unbounded domain, you'll only find one point, and you have to reason whether it's the max or min.



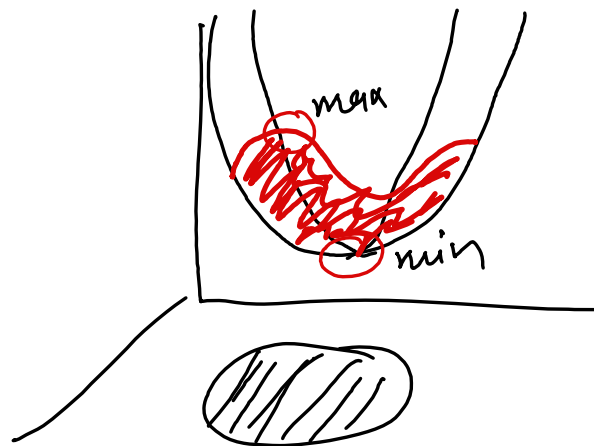
e.g. "optimize the distance from some point to some plane" you'll only get one answer — the closest point.

Think: there are infinitely many furthest points on the plane!
here: constraint is "on the plane" but that's an unbounded domain!



one equality
constraint

VS



closed bounded domain
constraint

e.g. Find pts at max/min for the f^n $f(x,y) = x - x^2 - y^2$
 over the set $D = \{(x,y) \in \mathbb{R}^2 : x^2 + 4y^2 \leq 1\}$ on and inside
an ellipse
 e.g. Find pts at max/min for the f^n $f(x,y,z) = 2x + y^2 - z - 4$
 over the set $D = \{(x,y,z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 \leq 1\}$ on and inside
a sphere.

Step 1: Find critical points of the f^u f lying inside the interior part of the domain D

e.g. find (a,b) s.t. $\nabla f(a,b) = (0,0)$ and, in our example,
 $a^2 + 4b^2 < 1$ (strictly less!)

Step 2: Find the points on the boundary (e.g. on $x^2 + 4y^2 = 1$) which may be points of max/min.
i.e. use method of Lagrange multipliers

Step 3: evaluate f of all the points we found in step 1 and step 2, to decide which lead to min/max.

[Notice, we don't use $D^2 f(a,b)$ at all! no second derivs!]

e.g. $f(x,y) = x - x^2 - y^2$ over $D = \{(x,y) \in \mathbb{R}^2 \mid x^2 + 4y^2 \leq 1\}$

Step 1 critical points s.t. $x^2 + 4y^2 < 1$:

we seek (x,y) s.t.
$$\begin{cases} f_x(x,y) = 0 \\ f_y(x,y) = 0 \\ x^2 + 4y^2 < 1 \end{cases} \Rightarrow \begin{cases} 1 - 2x = 0 \Rightarrow x = \frac{1}{2} \\ -2y = 0 \Rightarrow y = 0 \\ x^2 + 4y^2 < 1 \end{cases}$$

is $(\frac{1}{2}, 0)$ in the interior?

$(\frac{1}{2})^2 + 4(0)^2 = \frac{1}{4} < 1$ so $\boxed{(\frac{1}{2}, 0)}$ is a critical point.

Step 2 potential points of max/min on $x^2 + 4y^2 = 1$:

we seek $(x,y), \lambda$ s.t.
$$\begin{cases} f_x(x,y) = \lambda g_x(x,y) \\ f_y(x,y) = \lambda g_y(x,y) \\ g(x,y) = x^2 + 4y^2 - 1 = 0 \end{cases} \Rightarrow$$

$$\begin{cases} 1 - 2x = \lambda 2x \\ -2y = \lambda 8y \\ x^2 + 4y^2 - 1 = 0 \end{cases}$$

$$\begin{cases} 1-2x = \lambda 2x & \textcircled{1} \\ -2y = \lambda 8y & \textcircled{2} \\ x^2 + 4y^2 - 1 = 0 & \textcircled{3} \end{cases} \Rightarrow 8\lambda y + 2y = 0 \Rightarrow 2y(4\lambda + 1) = 0$$

$$\Rightarrow \underline{y=0} \text{ or } \lambda = -\frac{1}{4}$$

if $y=0$: use $\textcircled{3}$: $x^2 + 4(0)^2 - 1 = 0 \Rightarrow x^2 - 1 = 0 \Rightarrow x = \pm 1$

$$\boxed{(1, 0)}, \boxed{(-1, 0)}$$

check $\textcircled{1}$: $x=1$: $\textcircled{1} \Rightarrow 1 - 2(1) = 2\lambda(1) \Rightarrow -1 = 2\lambda \Rightarrow \lambda = -\frac{1}{2}$

important: no contradictions.

not important: λ -value for a specific pt

if $\lambda = -\frac{1}{4}$: $\textcircled{1} \Rightarrow 1 - 2x = -\frac{1}{4} \cdot 2x \Rightarrow 1 - 2x = -\frac{1}{2}x \Rightarrow 1 = -\frac{1}{2}x + 2x$

$$\Rightarrow 1 = \frac{3}{2}x \Rightarrow x = 1/3/2 = \frac{2}{3}, \quad x = \frac{2}{3}.$$

$$\textcircled{3} \Rightarrow \left(\frac{2}{3}\right)^2 + 4y^2 - 1 = 0 \Rightarrow \frac{4}{9} + 4y^2 - 1 = 0 \Rightarrow y^2 = \left(1 - \frac{4}{9}\right)/4$$

$$= (5/9)/4 = \frac{5}{36} \Rightarrow y = \pm \frac{\sqrt{5}}{6},$$

$$\boxed{\left(\frac{2}{3}, \frac{\sqrt{5}}{6}\right)}, \boxed{\left(\frac{2}{3}, -\frac{\sqrt{5}}{6}\right)}$$

Step 3: evaluate each pt and decide.

$$f = x - x^2 - y^2$$

$$\left(\frac{1}{2}, 0\right), (1, 0), (-1, 0), \left(\frac{2}{3}, \frac{\sqrt{5}}{6}\right), \left(\frac{2}{3}, -\frac{\sqrt{5}}{6}\right)$$

↓

$$f = \frac{1}{2} - \frac{1}{4} \\ = \frac{1}{4}$$

↑

biggest

↓

$$f = 0$$

↓

$$f = -1 - 1 \\ = -2$$

smallest

↓

$$f = \frac{2}{3} - \frac{4}{9} \\ - \frac{5}{36} \\ = \frac{1}{12}$$

↓

$$1/12$$

$\left(\frac{1}{2}, 0\right)$ is the point of maximum

$(-1, 0)$ is the point of minimum.

Bonus!

proof from here

Proof that $\nabla g(a,b) \perp (\text{tangent @ } (a,b))$ for $g(x,y) = k$, a level curve.

Suppose $g(x,y) = k$ and $k \in \mathbb{R}$, where $g: \mathbb{R}^2 \rightarrow \mathbb{R}$ is a continuously differentiable function. Let's name the curve formed by the set of points $(x,y) \in \mathbb{R}^2$ which satisfy $g(x,y) = k$, C_k . Consider a point on the curve, $(a,b) \in C_k$. Assume $\nabla g(a,b) \neq (0,0)$, and so, the curve is regular around (a,b) .

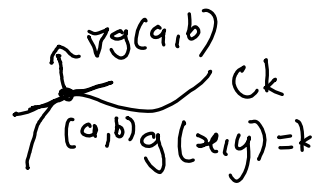
We'll describe that piece of curve around (a,b) by the vector $f^{\mathbb{R}} \vec{r}: (-\epsilon, \epsilon) \rightarrow \mathbb{R}^2$ with $\vec{r}(0) = (a,b)$ (note: the parametrization is arbitrary! We don't ever "use" that 0). Now, we know the tangent line to C_k at (a,b) will have direction $\vec{r}'(0)$! Let's see what else happens...

We know, around $(a,b) = \vec{r}(0)$, all the points outputted by $\vec{r}(t)$ (for $t \in (-\epsilon, \epsilon)$) will satisfy $g(\vec{r}(t)) = k$ (they're on the level curve!). Thus, differentiate both sides of $g(\vec{r}(t)) = k$ wrt t :

$$\frac{d}{dt}(g(\vec{r}(t))) = \frac{d}{dt} k \Rightarrow \nabla g(\vec{r}(t)) \cdot \vec{r}'(t) = 0 \quad \text{for every } t \in (-\epsilon, \epsilon).$$

when $t=0$ (recall $\vec{r}(0) = (a,b)$), we get

$$\nabla g(\vec{r}(0)) \cdot \vec{r}'(0) = 0 \Rightarrow \boxed{\nabla g(a,b) \cdot \vec{r}'(0) = 0}$$



Thus $\nabla g(a,b)$ is perpendicular to $\vec{r}'(0)$, which is the direction of the tangent line at $\vec{r}(0) = (a,b)$.

Theorem: Let $f, g: \mathbb{R}^2 \rightarrow \mathbb{R}$ be functions of class C^1 and let (a, b) be s.t. $g(a, b) = 0$ and $\nabla g(a, b) \neq (0, 0)$. If (a, b) is a solution to the problem "find the pts of min/max of $f(x, y)$ subject to $g(x, y) = 0$ ", then $\nabla f(a, b) \parallel \nabla g(a, b)$.

Proof:

Assume (a, b) is a point of minimum. The proof is the same for (a, b) a point of maximum.

g is continuously differentiable and $\nabla g(a, b) \neq (0, 0)$, so the level curve $g(x, y) = 0$ is regular around the point (a, b) , and we can represent that bit of regular curve by a vector f^n $\vec{r}: (-\epsilon, \epsilon) \rightarrow \mathbb{R}^2$ with $\vec{r}(0) = (a, b)$. Thus, as for any level curve, we know $\nabla g(a, b) \perp \vec{r}'(0)$.

Now, since $f(x, y)$ has a minimum at (a, b) , we can say, equivalently, that $f(\vec{r}(t))$ has a minimum at $t=0$ (since $\vec{r}(0) = (a, b)$). Thus, the derivative wrt t of $f(\vec{r}(t))$ at $t=0$ must be 0:

$$\frac{d}{dt}(f(\vec{r}(t))) = 0 \Rightarrow \nabla f(\vec{r}(0)) \cdot \vec{r}'(0) = 0 \Rightarrow \boxed{\nabla f(a, b) \cdot \vec{r}'(0) = 0}$$

so $\nabla f(a, b)$ and $\nabla g(a, b)$ are both perpendicular to $\vec{r}'(0)$, so $\nabla f(a, b) \parallel \nabla g(a, b)$.